

EXPERIMENT NO.6

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Roll No:3

Batch:A

Aim:Creating Smart Contracts and performing transactions using Solidity and Remix IDE

Theory:

A Smart Contract is a self-executing program stored on the blockchain that automatically enforces the terms of an agreement when predefined conditions are met. It eliminates the need for intermediaries, ensuring transparency, security, and trust between parties. Smart contracts are written in Solidity (for Ethereum-based systems) and executed on the Ethereum Virtual Machine (EVM).

Significance of Smart Contracts in the Ethereum Blockchain:

- **Automation:** Executes actions automatically when conditions are satisfied.
- **Transparency:** The contract code and transactions are visible on the blockchain.
- **Security:** Once deployed, contracts are immutable and tamper-proof.
- **Cost-effective:** Removes intermediaries and reduces transaction costs.
- **Trustless System:** Participants can interact without needing to trust each other directly.

Smart contracts form the backbone of decentralized applications (DApps) and decentralized finance (DeFi) systems on Ethereum.

Implementation:

The project “**Land Ownership Registry System**” focuses on automating land records and ownership transfers using Ethereum-based Smart Contracts.

To ensure the system works, we will implement two primary contracts:

1. **LandRegistry.sol:** Manages the database of lands and their owners.
2. **LandTransfer.sol:** Handles the logic of transferring ownership from a Seller to a Buyer.

Code:-

1. LandRegistry.sol

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

contract LandRegistry {
    struct Land {
        uint id;
        string location;
        uint area;
        address owner;
        bool isRegistered;
    }

    mapping(uint => Land) public lands;
    address public admin;

    constructor() {
        admin = msg.sender;
    }

    function registerLand(uint _id, string memory _location, uint _area, address _owner)
    public {
        require(msg.sender == admin, "Only admin can register land");
        require(!lands[_id].isRegistered, "Land already registered");

        lands[_id] = Land(_id, _location, _area, _owner, true);
    }

    function updateOwner(uint _id, address _newOwner) public {
        // This function will be called by the Transfer contract
        lands[_id].owner = _newOwner;
    }

    function getLand(uint _id) public view returns (uint, string memory, uint, address) {
        Land memory l = lands[_id];
```

```
        return (l.id, l.location, l.area, l.owner);
    }
}
```

2. LandTransfer.sol

```
// SPDX-License-Identifier: MIT
```

```
pragma solidity ^0.8.0;
```

```
import "./LandRegistry.sol";
```

```
contract LandTransfer {
    LandRegistry public registry;
```

```
    constructor(address _registryAddress) {
        registry = LandRegistry(_registryAddress);
    }
```

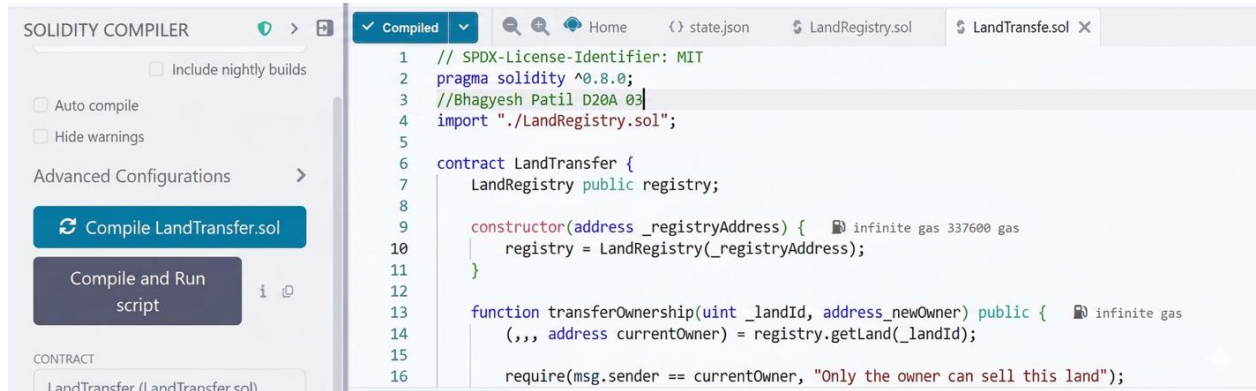
```
    function transferOwnership(uint _landId, address _newOwner) public {
        (,, address currentOwner) = registry.getLand(_landId);
```

```
        require(msg.sender == currentOwner, "Only the owner can sell this land");
```

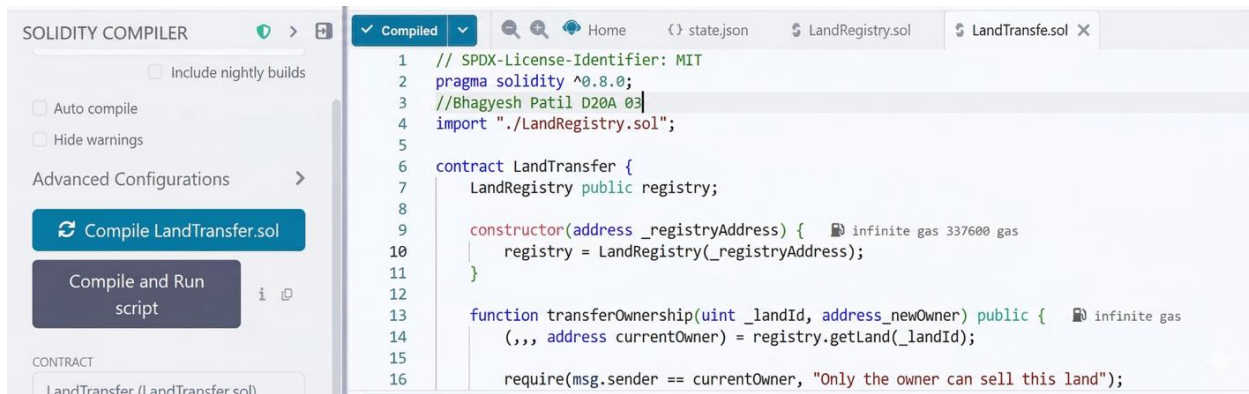
```
        registry.updateOwner(_landId, _newOwner);
    }
}
```

Step 1: Set up the Files

1. Go to Remix IDE.
2. In the File Explorer, create a new folder named LandRegistryProject.
3. Create two files: LandRegistry.sol and LandTransfer.sol.



Step 2: Compile both the contracts



Step 3: Deploy the LandRegistry contract

The screenshot displays the Remix IDE interface during the deployment of the LandRegistry contract. The left sidebar shows the 'TRANSACTIONS' panel with the following settings:

- Address: 0x5B3...eddC4 (99.9999999999)
- Gas Limit: Estimated Gas (3000000)
- Value: 0 Wei
- Contract: LandRegistry - LandRegistryProject/L
- Deploy button is highlighted.

The main area shows the Solidity code for the LandRegistry contract:

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.0;
3 //Bhagyesh Patil D20A 03
4 import "./LandRegistry.sol";
5
6 contract LandTransfer {
7     LandRegistry public registry;
8
9     constructor(address _registryAddress) {
10         registry = LandRegistry(_registryAddress);
11     }
12
13     function transferOwnership(uint _landId, address _newOwner) public {
14         (,,, address currentOwner) = registry.getLand(_landId);
15         require(msg.sender == currentOwner, "Only the owner can sell this land");
16     }
17 }
```

Below the code, the 'Explain contract' button is visible. The bottom panel shows the deployment log:

```
[vm] from: 0x5B3...eddC4 to: LandRegistry.(constructor) value: 0 wei data: 0x608...f0033 logs: 0 hash: 0xb56...
```

The log indicates a successful transaction with the following details:

- status: 1 Transaction mined and execution succeed
- transaction hash: 0xb56f7e907d2d8325540082d1bb3d3cd6c26577adbb830c9d156fa2c4c83ac60
- block hash: 0x97cd49221691d65cb4aa5e98bbf679bffa4debea6cf34669b25e943463de3ab
- block number: 124
- contract address: 0xc611d78A9287DfdF85faB1c22135977721dd96f4c
- from: 0x5838da6a701c568545dcfc807Fc8878f56beddc4

Step 4: Copy the contract address of the deployed LandRegistry contract

15	
16	<code>require(msg.sender == currentOwner, "Only the owner can sell this land");</code>
 Explain contract	
0 ☐ Listen on all transactions 	
transaction hash	0xb50f7e067d2d9329540023d1b65d9cd6c265f7adbb030c9d156fa2c4c83ac60 
block hash	0x97cd49221691d65cb4ea5e98bbf6798ffa4debeea6cf346b9b25e943463de3ab 
block number	124 
contract address	0xC61d78A92B7DfdF85fAB1c22135977721dd96F4c 
from	0x58380a6a701c568545dCfcB03FcB875f56beddC4 
to	LandRegistry.(constructor) 
transaction cost	824515 gas 

Step 5: Select LandTransfer here in the dropdown

GAS LIMIT

☒ Estimated Gas

☐ Custom 3000000

VALUE

Wei

CONTRACT

LandTransfer - LandRegistryProject/Li

LandRegistry - LandRegistryProject/LandRegistry.sol

LandTransfer - LandRegistryProject/LandTransfer.sol

At Address

Load contract from Address

Transactions recorded 2

>

Deployed Contracts 2

>

> LANDREGISTRY AT 0XC61...96F4   

> LANDTRANSFER AT 0XA2E...DC1   

4 import "../LandRegistry.sol";

5

6 contract LandTransfer {

7 LandRegistry public registry;

8

9 constructor(address _registryAddress) {  infinite gas 337600 gas

10 | registry = LandRegistry(_registryAddress);

11 }

12

13 function transferOwnership(uint _landId, address _newOwner) public {  infinite gas

currentOwner) = registry.getLand(_landId);

sender == currentOwner, "Only the owner can sell this land");

creation of LandRegistry pending...

 [vm] from: 0x583...eddC4 to: LandRegistry.(constructor) value: 0 wei data: 0x608...f0033 logs: 0 hash: 0xb50...3ac60

status 1 Transaction mined and execution succeed

transaction hash 0xb50f7e067d2d9329540023d1b65d9cd6c265f7adbb030c9d156fa2c4c83ac60 

block hash 0x97cd49221691d65cb4ea5e98bbf6798ffa4debeea6cf346b9b25e943463de3ab 

block number 124 

contract address 0xC61d78A92B7DfdF85fAB1c22135977721dd96F4c 

0 ☐ Listen on all transactions 

Filter with

Step 6: Add the contract address in place of _registryAddress field next to the deploy button, and click on deploy so as to deploy the LandTransfer contract

DEPLOY & RUN
TRANSACTIONS

+ Authorize Delegation

GAS LIMIT

☒ Estimated Gas

☐ Custom

3000000

VALUE

0

Wei

CONTRACT

LandTransfer - LandRegistryProject/Li

evm version: osaka

Deploy

0xC61d78A92B7DfdF85fAB1c22

▼

At Address

Load contract from Address

✓

[vm] from: 0x583...eddC4 to: LandTransfer.(constructor) value: 0 wei data: 0x608...96f4c logs: 0 hash: 0x02b...545c1

Debu

status

1 Transaction mined and execution succeed

transaction hash

0x02bfc4bb38722b91c6980f77829c9fe52aa29f2f6e96765034aa9b1a3f4545c1

block hash

0x30d868035e979b6362cde68e0892bca6d6a6f80774dc8473ada8cb5da89296bb

block number

125

contract address

0xa2e9669fc58d05500aF18aEd20dcF10A9e00Cb97

from

0x5838Da6a701c568545dCfcB03FcB875f56beddC4

to

LandTransfer.(constructor)

transaction cost

443826 gas

execution cost

360488 gas

output

0xf08060a0c73d8015c1000fc75f5f45b50600d361061003d575f3560a01c00630507f731d6100385780c37b1030001d61005d575b5f5ff45b610057600d8036030

Step 7: Execute the transactions

1.Land Registration

In the deployed contracts section expand the LandRegistry contract, and add details as mentioned. Also ensure that the wallet address is added in the owner field. Once all this is done, click on transact to complete the transaction.

Deployed Contracts 2

LANDREGISTRY AT 0XC61...96F

Balance: 0 ETH

REGISTERLAND

_id: 1

_location: "Mumbai"

_area: 500

_owner: 0x5B38Da6a701c568545dCfcB03FcI

Calldata

Parameters

transact

DEPLOY & RUN TRANSACTIONS



ENVIRONMENT  

 Reset Sta

Remix VM (Osaka) 

VM

ACCOUNT    

0x5B3...eddC4 (99.9999999999 

0x5B3...eddC4 (99.999999999999746

0xAb8...35cb2 (100.0 ETH)

0x4B2...C02db (100.0 ETH)

0x787...cabaB (100.0 ETH)

0x617...5E7f2 (100.0 ETH)

0x17F...8c372 (100.0 ETH)

0x5c6...21678 (100.0 ETH)

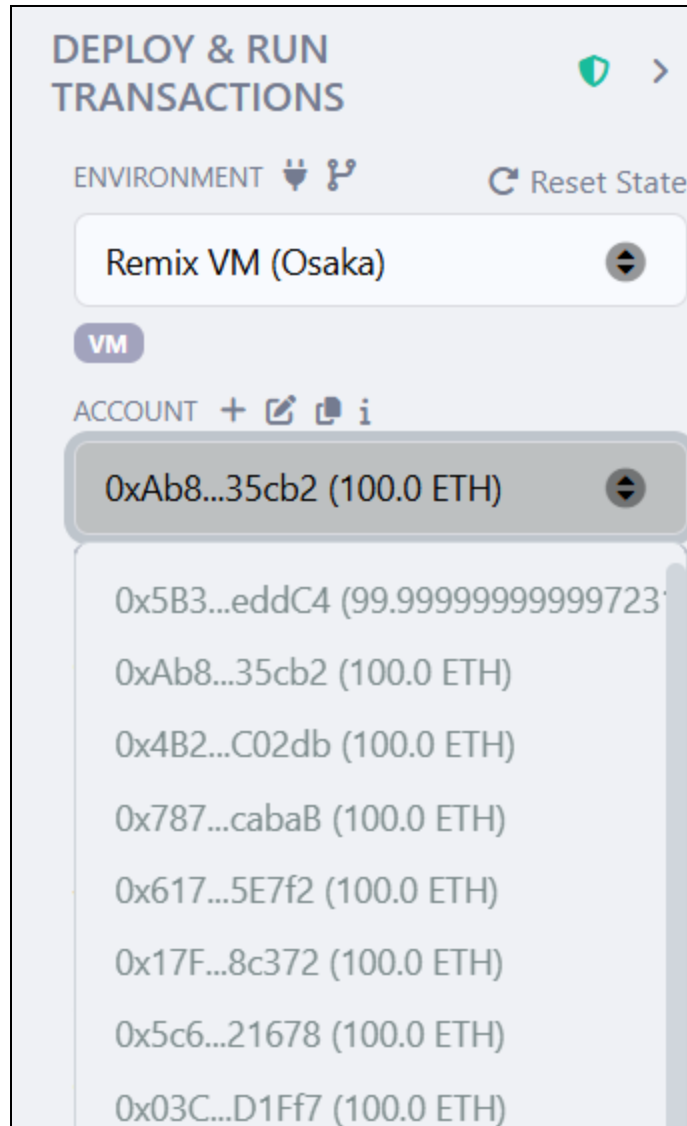
0x03C...D1Ff7 (100.0 ETH)

0x1aE...E454C (100.0 ETH)

0x0A0...C70DC (100.0 ETH)

2. Transfer Ownership

Copy a different wallet address from the account dropdown (this wallet address represents the buyer)



Switch back to the original wallet address(i.e the seller's wallet address) and within the transferOwnership function in the land transfer contract, enter:-

_landId: **1**

_newOwner: **[The Buyer's Address]**

✓ LANDTRANSFER AT 0XA2E...DC

✕

Balance: 0 ETH

TRANSFEROWNERSHIP

_landId:

1

_newOwner:

0xAb8483F64d9C6d1EcF9b849Ae67

Calldata

Parameters

transact

registry

Click on transact to complete the ownership transfer

0 ☐ Listen on all transactions🔍

transact to LandTransfer.transferOwnership pending ...

✓

[vm] from: 0x583...eddC4 to: LandTransfer.transferOwnership(uint256,address) 0xa2e...DCb97 value: 0 wei data: 0x295...35cb2 logs: 0
hash: 0xc18...d9a2b

Debug

status	1 Transaction mined and execution succeed
transaction hash	0xc188c22aff87cc55a3c65af0a5b87c469e02389c574b8ee4eb3c844087ad9a2b 🔗
block hash	0x3c0030a59908a64701260b4e6b9ead20a246afe02d20af27f0bbe6081d7bbe0d 🔗
block number	127 🔗
from	0x5838Da6a701c568545dCfcB03FcB875f56beddC4 🔗
to	LandTransfer.transferOwnership(uint256,address) 0xa2e9669fC58d055D0aF18aEd20dcF10A9e0DCb97 🔗
transaction cost	43957 gas 🔗
execution cost	22385 gas 🔗
output	0x 🔗
decoded input	{ "uint256 landId": "1",

Now, verify the ownership transfer by going to the LandRegistry contract. When we click on the getLand button, we see a different wallet address, indicating that the ownership transfer has taken place and is successfully verified.

Deployed Contracts

REGISTERLAND
⬆

_id:

_location:

_area:

_owner:

Calldata
Parameters
transaction

updateOwner

uint256_id, address_newO

admin

getLand

1

0: uint256: 1

1: string: Mumbai

2: uint256: 500

3: address: 0xABB483F64d9C6d1EcF9b849Ae677dD3315835cb2

LandRegistry.sol
LandTransfer.sol

```

1  pragma solidity ^0.8.0;
2  //Bhagyesh Patil D20A 03
3  import "./LandRegistry.sol";
4
5
6  contract LandTransfer {

```

✖ Explain contract

transaction cost
43957 gas

execution cost
22385 gas

output
0x

decoded input

```

{
  "uint256_landId": "1",
  "address_newOwner": "0xab483f64d9c6d1ecf9849ae677dd3315835cb2"
}

```

decoded output
()

logs
[]

raw logs
[]

call to LandRegistry.getLand

CALL [call] from: 0x5838Da6a701c568545dCfc803Fc8875f56beddC4 to: LandRegistry.getLand(uint256) data: 0xf02.

Conclusion:-

This experiment helped in understanding how smart contracts automate processes and execute securely on the Ethereum blockchain. Using Solidity and Remix IDE, the development, deployment, and interaction with smart contracts become simple and efficient, forming the foundation for decentralized applications.